BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter Part-I, 2023 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. In collinator the rays coming out of the lens are parallel if the distance between lens and slit is:

(A) Equal to the focal length of lens
(B) Greater than the focal length of lens
(C) Less thanthe focal length of lens
(D) At any distance between lens and slit

Q2. The moment of inertia of a cylinder is:

(A) (2/5) mr^2
(B) (1/2) mr^2
(C) mr^2
(D) (1/12) mr^2

Q3. Slope of velocity-time graph gives:

(A) Acceleration
(B) Distance
(C) Momentum
(D) Displacement

Q4. If a wave travelling indenser medium is reflected from the boundary of rarer medium, the phase change in the wave is:

(A) 0 degrees
(B) 180 degrees
(C) 90 degrees
(D) 60 degrees

Q5. 1 torr = Nm^-2 :

(A) 133.3
(B) 143.3
(C) 153.3
(D) 123.3

Q6. In measurement 8,000 kg, if the scale has a least count of 10kg, then the number of significant figures are

(A) 3
(B) 4
(C) 2
(D) 1

Q7. An ideal reversible heat engine has efficiency:

(A) 100%
(B) Highest
(C) Efficiency depends upon the nature of working substance
(D) 33.33%

Q8. The angle between velocity and acceleration at the highest point during the projectile motion is:

(A) 45 degrees
(B) 30 degrees
(C) 90 degrees
(D) 0 degrees

Q9. In Michelson's interferometer, to observe two consecutive dark and bright fringes, themovable mirror is moved through:

(A) lambda/2
(B) lambda/4
(C) lambda/3
(D) lambda/[illegible in source scan]

Q10. Which of the following is supplementary unit:

(A) Ampere
(B) Candela
(C) Mole
(D) Steradian

Q11. The angle theta = omega specifies in SHM:

(A) Displacement
(B) Direction of motion of the point
(C) Both displacement and direction
(D) Direction of force

Q12. If the magnitude of resultant of two equal forces is also equal to the magnitude of either forces, then the angle between forces is:

(A) 45 degrees
(B) 120 degrees
(C) 90 degrees
(D) 60 degrees

Q13. If the source and sink are the same temperature, then net change in entropy is:

(A) Minimum
(B) Maximum
(C) Zero
(D) Negative

Q14. According to Doppler effect, a star moving towards the earth show:

(A) Red shift
(B) Blue shift
(C) Yellow shift
(D) Green shift

Q15. If the frequency of rotation of a spacecraft is doubled, then gravity produced becomes:

(A) Double
(B) 3 times
(C) 4 times
(D) Does not change

Q16. As the food we eat in the day has about the same energy as:

(A) (1/2) litre of petrol
(B) (1/3) litre of petrol
(C) (1/50) litre of petrol
(D) (1/[illegible in source scan]) litre of petrol

Q17. For [cut off at bottom of source scan]

(A) [cut off at bottom of source scan]
(B) [cut off at bottom of source scan]
(C) [cut off at bottom of source scan]
(D) [cut off at bottom of source scan]

Essay Type

Time Allowed: 2.40 Hrs. | **Maximum Marks:** 68

QQ.2. (Marks 16) Write short answers to any Eight (8) questions.

i Write two dimensions of: (i) Co-efficient of viscosity (ii) Pressure

ii A light year is the distance travelled by light in one year. How many meteres are there in one light year? (Speed of light = 3 x 10^8 m/s)

iii What is an error? Write down its types.

iv How the uncertainty is reduced in a timing experiment?

v Two vectors have un-equal magnitudes. Can their sum be zero? Explain.

vi Under what circumstances would a vector have components that are equal in magnitude?

vii A bar and B bar are two non-zero vectors. How can their scalar product be zero? How can their vector product be zero?

viii At what point or points in its path does a projectile have its minimum speed, its maximum speed?

ix As an object is thrown vertically upwards, its velocity decreases, is this against the law of conservation of linear momentum?

x If H is height attained by a projectile and T is the time of flight, then H = gT^2/8

xi What is impulse? How it is related to momentum?

xii A person is standing near a fast moving train. Is there any danger that he will fall towards it?

QQ.3. (Marks 16) Write short answers to any Eight (8) questions.

i Calculate the work done when a 50kg bag of books is lifted through 50cm.

ii When a rocket re-enters the atmosphere, its nose cone becomes veryhot? Where does this heat energy come from?

iii Discuss the relation and the importance of -ve sign in the relation U_g = GMm/r

iv How centripetal force acts and give two forces which can provide centripetal force to the circulating system?

v How would you explain the concept of moment of inertia in orbital and spin angular momentum?

vi Explain how many minimum number of geo-stationary satellites are required for global coverage of T.V. transmission?

vii What should be the length of simple pendulum whose period is 2 sec?

viii Does the acceleration of a simple harmonic oscillator remain constant during its motion? Is acceleration ever zero? Explain.

ix What is meant by phase angle? Does it define angle between maximum displacement and the driving force?

x Why Newton's formula of speed of sound has 16% error? Support your answer by proper reasoning.

xi How beats are useful in tuning musical instruments?

xii Define the terms crest and trough.

QQ.4. (Marks 12) Write short answers to any Six (6) questions.

i Can visible light produce interference fringes? Explain.

ii How would you distinguish between un-polarized and plane-polarized light?

iii Define air film. Write its two examples.

iv Why would it be advantageous to use blue light with a compound microscope?

v What are the uses of spectrometer?

vi Why does the pressure of a gas in a car tyre increases when it is driven through some distance?

vii Is it possible to construct a heat engine that will not expel heat into the atmosphere?

viii Derive Charles' law from kinetic theory of gases.

ix Define triple point of water, also write its value.

QSECTION-II. Note: Attempt any THREE questions.

Q5.(a). (Marks 5) Define vector.

Q5.(b). (Marks 3) How large force is required to accelerate an electron (m = 9.1 x 10^-31 kg) from rest to a speed of 2 x 10^7 ms^-1 through a distance of 5.0cm?

Q6.(a). (Marks 5) What are geostationary orbits and geostationary satellites? Derive the relation for the radius of a geostationary satellite.

Q6.(b). (Marks 3) A football is throwun upward with an angle of 30 degrees with respect to the horizontal. To throw a 40m pass what must be the initial speed of the ball?

Q7.(a). (Marks 5) State and prove the Bernoulli's equation in fluid dynamics that relates the pressure to the fluid speed and height.

Q7.(b). (Marks 3) A heat engine performs 100J of work and at the same time rejects 400J of heat energy to the cold reservoirs. What is the efficiency of the engine?

Q8.(a). (Marks 5) What is simple pendulum? Show that its motion is SHM. Derive an expression for its time period.

Q8.(b). (Marks 3) The wavelength of the signal from a radio transmitter is 1500m and frequency is 200 KHz. What is the wavelength for a transmitter operating at 1000KHz and with what speed of the radio waves travel?

Q9.(a). (Marks 5) What is diffraction of light? Calculate the wavelength of light used by diffraction grating.

Q9.(b). (Marks 3) A simple astronomical telescope in normal adjustment has an objective of focal length 100cm and an eye piece of focal length 5.0cm. Calculate the angular magnification.

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: Inter, Part I, 2023 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The branch of physics which is concerned with ulti..... matter is composed is:

(A) Atomic Physics
(B) Nuclear Physics
(C) Plasma Physics
(D) Particle physics

Q2. Time interval between normal heart beats is:

(A) 1x10^-6 s
(B) 8x10^1 s
(C) 1x10^-2 s
(D) 8x10^-1 s

Q3. If A bar = 4i, B bar = 6j then A bar x B bar will be

(A) 24k
(B) 24
(C) Zero
(D) -24k

Q4. If the magnitude of scalar and vector product of two vectors are 2 root(3) and 2 respectively, the angle between vectors is:

(A) 30 degrees
(B) 45 degrees
(C) 120 degrees
(D) 180 degrees

Q5. The rate of change of momentum of a body falling freely is equal to its:

(A) K.E.
(B) Weight
(C) Power
(D) Momentum

Q6. A typical rocket eject the burn gases at speeds of over:

(A) 400 m/sec
(B) 4000 m/sec
(C) 8000 m/sec
(D) 10000 m/sec

Q7. Two masses of 1 gm and 4 gm are moving with same K.E. The ratio of their linear momentum are:

(A) 1 : 16
(B) 6 : 1
(C) 1 : 2
(D) 4 : 1

Q8. When a body moves in a circle its linear velocity always:

(A) Remains constant
(B) Becomes zero
(C) Changes
(D) Increases

Q9. The rotational K.E. of Disc is equal to:

(A) (1/4) mv^2
(B) (1/2) mv^2
(C) (1/2) I omega
(D) I omega

Q10. Viscosity of air at 30 degrees C is:

(A) 0.019 x 10^-3 Nsm^-2
(B) 0.295 x 10^-3 Nsm^-2
(C) 0.510 x 10^-3 Nsm^-2
(D) 0.564 x 10^-3 Nsm^2

Q11. A heavily damped system has a fairly flat resonance curve in an :

(A) Velocity time graph
(B) Distance time graph
(C) Amplitude frequency graph
(D) Amplitude time graph

Q12. Bats navigate and find food by:

(A) Microwaves
(B) Echo location
(C) Electromagnetic waves
(D) Matter waves

Q13. Longitudinal waves do not exhibit:

(A) Reflection
(B) Refraction
(C) Polarization
(D) Diffraction

Q14. The first demonstration of wave nature of light was provided in 1801 by the experiment of:

(A) Huygen
(B) Newton
(C) Maxwell
(D) Thomas Young

Q15. The diameter of the core of multimode step index fibre is:

(A) 50 mm
(B) 50 um
(C) 50 nm
(D) 50 Pm

Q16. If temperature of a sink of a heat engine is absolute zero, the efficiency of heat engine should be:

(A) 100%
(B) 50%
(C) 0%
(D) Infinity

Q17. The ratio of C_p/C_v for a diatomic gas is equal to:

(A) 1.67
(B) 1.50
(C) 1.40
(D) 1.29

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Give five instruments for indicating units.

ii What is random error. How it can be eliminated?

iii Why we find it useful to have two units for the amount of substance, kilogram and mole?

iv Does a dimensional analysis give any information on constant of proportionality that may appear in an algebraic expression? Explain.

v How would you verify that the dot and cross product become equal to magnitude?

vi If all the components of the vectors A1 bar and A2 bar were reversed, how would this alter A1 bar x A2 bar.

vii Name three different conditions that could make A1 bar x A2 bar = O bar

viii Does a moving object has impulse? Explain your reasoning.

ix A 1500 kg car has its velocity reduced from 20 m/s to 15 m/s in 3.0 second. How large was the average retarding force?

x Define impulse and show that how it is related to linear momentum?

xi Why isolated system is important to conserve linear momentum? Also state law of conservation of momentum.

xii Why fog droplets appear to be suspended in air?

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i When a rocket re-enters the atmosphere, its nose cone becomes very hot? Where does this heat energy come from?

ii A boy uses a catapult to throw a stone which accidentally smashes a green house window. List the possible energy changes.

iii State work-energy principle.

iv What is meant by moment of inertia? Explain its significance.

v Explain how many minimum number of geo-stationary satellites are required for global coverage of T.V. transmission?

vi Differentiate between tangential velocity and angular velocity.

vii What happens to the period of simple pendulum, if its length is doubled? What happens if the suspended mass is doubled?

viii What is meant by phase angle? Does it define angle between maximum displacement and the driving force?

ix Differentiate between transverse waves and longitudinal waves.

x What should be the frequency of a simple pendulum whose period is 0.5 seconds at a place where g = 9.8 m/s^-2?

xi A wave is produced along a stretched string but some of its particles permanent show zero displacement. What type of wave is it?

xii Explain why sound travels faster in warm air than in cold war?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Under what conditions two or more sources of light behave as coherent sources?

ii How would you manage to get more orders of spectra using a diffraction grating?

iii Draw an interference pattern formed with white light.

iv Why would it be advantageous to use blue light with a compound microscope?

v Draw ray diagram of compound microscope and write its total magnification.

vi Is it possible to construct a heat engine that will not expel heat into the atmosphere?

vii Can mechanical energy be converted completely into heat energy? If so give an example.

viii Derive the Charles's law from kinetic theory of gases.

ix What is adiabatic process? Write down its two examples.

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 5) Define Rectangular component. Explain addition of vectors by rectangular components.

QQ.5(b). Ten bricks, each 6.0 cm thick and mass 1.5 kg lie flat on a table. How much work is required to stack them one on the top of another?

QQ.6(a). Define projectile motion. Derive the expression for: (i) Time of flight and (ii) Height of a projectile

QQ.6(b). What is the least speed at which an aeroplane can execute a vertical loop of 1.0 km radius so that there will be no tendency for the pilot to fall down at the highest point.

QQ.7(a). Show that the pressure exerted by the gas is directly proportional to the average translational kinetic energy of gas molecules.

QQ.7(b). Water flows through a hose, whose internal diameter is 1 cm at a speed of 1ms^-1. What should be the diameter of the nozzle if the water is to merge at 2 ms^-1?

QQ.8(a). Describe Doppler's Effect? Derive apparent frequency and discuss its results if (i) Source is moving towards the stationary observer (ii) Source is moving away from stationary observer

QQ.8(b). Find the amplitude and frequency of an object vibrating at the end of a spring, if the equation for its position, as function of time is X = 0.25cos(pi/g)t

QQ.9(a). Describe the principle, construction and working of Michelson's interferometer. How can you find the wavelength of light used?

QQ.9(b). A compound microscope has lenses of focal length 1.0 cm and 3.0 cm. An object is placed 1.2 cm from the object lens. If a virtual image is formed, 25 cm from

the eye, calculate the separation of the lenses and the magnification of the instrument.